Student Name: ___________Wajiha Nazir________________Roll No: _____25K-3021___________ Section: BSE-1B

Question:1

Write a C program that takes 5 integers in an array and shifts all elements one position to the right, and moves the last element to the first position.

Example:

Input → 1 2 3 4 5

Output → 5 1 2 3 4

```c
#include <stdio.h>

int main()
{
    int arr[5], i, temp;
    for(i = 0; i < 5; i++)
    {
        printf("Enter number %d: ", i + 1);
        scanf("%d", &arr[i]);
    }
    temp = arr[4];
    for(i = 4; i > 0; i--)
    {
        arr[i] = arr[i - 1];
    }
    arr[0] = temp;
    for(i = 0; i < 5; i++)
    {
        printf("%d", arr[i]);
    }
    return 0;
}
```

```
Enter number 1: 1
Enter number 2: 2
Enter number 3: 3
Enter number 4: 4
Enter number 5: 5
51234
--------------------------------
Process exited after 3.595 seconds with return value 0
Press any key to continue . . .
```

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Question:2

Write a program that reads the 10 numbers from user and store these numbers into an array of same size. You program should provide a searching mechanism in such a way that how many times a particular number occurred and then print it on screen. If number is not in array, then program should display a message “number not found”.

```c
#include<stdio.h>

int main()
{
    int arr[10], i, num, count=0;
    for(i=0;i<10;i++)
    {
        printf("Enter Number %d:\t",i+1);
        scanf("%d",&arr[i]);
    }
    printf("Enter the Number You Want To Search:\t");
    scanf("%d",&num);
    for(i=0;i<10;i++)
    {
        if(arr[i]==num)
        {
            count++;
        }
    }
    if(count>0)
        printf("You Entered %d Times Number %d",count,num);
    else
        printf("Number Not Found");

    return 0;
}
```

```
Enter Number 1: 2
Enter Number 2: 5
Enter Number 3: 4
Enter Number 4: 9
Enter Number 5: 7
Enter Number 6: 6
Enter Number 7: 5
Enter Number 8: 4
Enter Number 9: 2
Enter Number 10:        1
Enter the Number You Want To Search:    2
You Entered 2 Times Number 2
--------------------------------
Process exited after 16.81 seconds with return value 0
Press any key to continue . . .
```

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```
Enter Number 1: 5
Enter Number 2: 4
Enter Number 3: 8
Enter Number 4: 9
Enter Number 5: 1
Enter Number 6: 25
Enter Number 7: 4
Enter Number 8: 5
Enter Number 9: 6
Enter Number 10:        8
Enter the Number You Want To Search:    7
Number Not Found
--------------------------------
Process exited after 12.73 seconds with return value 0
Press any key to continue . . .
```

Question:3

Sir Saif ask you to write a program which can help him in storing your quiz marks for pass students within range [5-10] will be stored, consider there are 10 students registered in Section BSE-1B. He wants an array of same size where marks for failed students within range [0-5] are stored. Write a program for the given scenario. You program should exit if user enters -1 and fill display the marks entered along with average of each array.

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```c
#include <stdio.h>

int main()
{
    int mark, i;
    int pass[10], fail[10], passCount = 0, failCount = 0;
    float passSum = 0, failSum = 0;
    printf("Enter quiz marks for students (enter -1 to stop):\n");
    for(i = 0; i < 10; i++)
    {
        printf("Enter mark for student %d: ", i + 1);
        scanf("%d", &mark);
        if(mark == -1)
            break;
        if(mark >= 5 && mark <= 10)
        {
            pass[passCount++] = mark;
            passSum += mark;
        }
        else if(mark >= 0 && mark < 5)
        {
            fail[failCount++] = mark;
            failSum += mark;
        }
        else
        {
            printf("Invalid mark! Please enter between 0 and 10.\n");
            i--;
        }
    }
    printf("\nMarks of Passed Students (5-10): ");
    if(passCount == 0)
        printf("None");
    else
        for(i = 0; i < passCount; i++)
            printf("%d ", pass[i]);

    printf("\nMarks of Failed Students (0-4): ");
    if(failCount == 0)
        printf("None");
    else
        for(i = 0; i < failCount; i++)
            printf("%d ", fail[i]);
    if(passCount > 0)
        printf("\n\nAverage of Passed Students = %.2f", passSum / passCount);
    else
        printf("\n\nNo passed student marks entered.");

    if(failCount > 0)
        printf("\nAverage of Failed Students = %.2f\n", failSum / failCount);
    else
        printf("\nNo failed student marks entered.\n");
    return 0;
}
```

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```
Enter quiz marks for students (enter -1 to stop):
Enter mark for student 1: 2
Enter mark for student 2: 8
Enter mark for student 3: 9
Enter mark for student 4: 4
Enter mark for student 5: 6
Enter mark for student 6: 3
Enter mark for student 7: 7
Enter mark for student 8: 1
Enter mark for student 9: 5
Enter mark for student 10: -1

Marks of Passed Students (5-10): 8 9 6 7 5
Marks of Failed Students (0-4): 2 4 3 1

Average of Passed Students = 7.00
Average of Failed Students = 2.50

--------------------------------
Process exited after 18.48 seconds with return value 0
Press any key to continue . . .
```

Question:4

Your class teacher asks you to develop a program that can help in converting mixed-case messages entered by users. The program should read a sentence using scanf(”%[^\n]”, str); and then convert all lowercase letters to uppercase and all uppercase letters to lowercase, without using any string library functions. Finally, display the converted message back to the user. Example: Input → HeLLo WoRLd → Output → hEllO wOrlD

```c
#include <stdio.h>

int main()
{
    char str[100];
    int i;
    printf("Enter a sentence: ");
    scanf("%[^\n]", str);
    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'A' && str[i] <= 'Z')
            str[i] = str[i] + 32;
        else if(str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
    }
    printf("\nConverted message: %s\n", str);
    return 0;
}
```

```
Enter a sentence: HeLLo WoRLd

Converted message: hEllO wOrlD

--------------------------------
Process exited after 30.37 seconds with return value 0
Press any key to continue . . .
```

Question:5

Write a program that reads 10 integers into an array. Finds and prints the difference (max - min) between the largest and smallest elements.

```c
#include<stdio.h>

int main(){
	int arr[10],i,max=0,min=0,diff;
	for(i=0;i<10;i++)
	{
		printf("Enter %d Number:\t",i+1);
		scanf("%d",&arr[i]);
		max=arr[0];
		min=arr[0];
	}
	for(i=0;i<10;i++)
	{
		if(max<arr[i])
			max=arr[i];
		else
			min=arr[i];
	}
		printf("Maximum is  %d",max);
	printf("\nMinimum is %d",min);
	diff=max-min;
	
	printf("\nDifference is %d",diff);
	return 0;
}
```

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```
Enter 1 Number: 1
Enter 2 Number: 63
Enter 3 Number: 1
Enter 4 Number: 9
Enter 5 Number: 85
Enter 6 Number: 2
Enter 7 Number: 3
Enter 8 Number: 6
Enter 9 Number: 4
Enter 10 Number:        1
Maximum is  85
Minimum is 1
Difference is 84
--------------------------------
Process exited after 8.225 seconds with return value 0
Press any key to continue . . .
```

Question: 6

You are assisting your English language teacher who wants to analyze how many vowels and consonants appear in students' submitted words. Write a program that reads a single word using scanf("%s", str); and counts the number of vowels and consonants in it without using any string library functions like strlen(). The program should display both counts on the screen.

```c
#include<stdio.h>

int main(){
    int num, i, vowels=0, cons=0;
    printf("Enter how many times you want to enter characters:\t");
    scanf("%d",&num);
    char words[num];
    for(i=1;i<=num;i++)
    {
        printf("Enter Character %d:\t",i);
        scanf("%s",&words[i]);
        words[i]=tolower(words[i]);
        if(words[i]=='a' || words[i]=='e' || words[i]=='i' || words[i]=='o' || words[i]=='u')
            vowels++;
        else
            cons++;
    }
    printf("You Entered %d vowels",vowels);
    printf("\nYou Entered %d consonants",cons);
    return 0;
}
```

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```
Enter how many times you want to enter characters:      10
Enter Character 1:      A
Enter Character 2:      k
Enter Character 3:      o
Enter Character 4:      N
Enter Character 5:      R
Enter Character 6:      O
Enter Character 7:      Q
Enter Character 8:      U
Enter Character 9:      m
Enter Character 10:     D
You Entered 4 vowels
You Entered 6 consonants
--------------------------------
Process exited after 30.28 seconds with return value 0
Press any key to continue . . .
```

Question:7

You are working in a data management department where you are asked to remove duplicate entries from a list of recorded IDs. Write a program that takes 10 integers from the user and stores them in an array. The program should replace every duplicate element (after its first occurrence) with -1 to mark it as a duplicate and then display the final updated array on the screen.

Example: Input → 2 3 5 3 5 9 1 2 8 1 → Output → 2 3 5 -1 -1 9 1 -1 8 -1

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```c
#include <stdio.h>

int main()
{
    int arr[10];
    int i, j;
    for(i = 0; i < 10; i++)
    {
        printf("Enter ID %d:\t",i+1);
        scanf("%d", &arr[i]);
    }
    for(i = 0; i < 10; i++)
    {
        for(j = i + 1; j < 10; j++)
        {
            if(arr[i] == arr[j])
                arr[j] = -1;
        }
    }
    printf("\nUpdated IDs (duplicates marked as -1):\n");
    for(i = 0; i < 10; i++)
        printf("%d ", arr[i]);
    printf("\n");
    return 0;
}
```

```
Enter ID 1:     2
Enter ID 2:     3
Enter ID 3:     5
Enter ID 4:     3
Enter ID 5:     5
Enter ID 6:     9
Enter ID 7:     1
Enter ID 8:     2
Enter ID 9:     8
Enter ID 10:    1

Updated IDs (duplicates marked as -1):
2 3 5 -1 -1 9 1 -1 8 -1

--------------------------------
Process exited after 22.01 seconds with return value 0
Press any key to continue . . .
```

Question:8

Write a program that reads a string using scanf("%[^A-Za-z]", str); so that it accepts and stores all characters except alphabets. The program should then display the entered non-alphabetic characters on the screen.

```c
#include <stdio.h>

int main() {
    int num, i, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    for (i = 1; i < num; i++) {
        if (num % i == 0) {
            sum = sum + i;
        }
    }
    if (sum == num)
        printf("%d is a Perfect Number.", num);
    else
        printf("%d is NOT a Perfect Number.", num);
    return 0;
}
```

```
Enter a string: 1234wajiha
You entered: 1234

--------------------------------
Process exited after 4.282 seconds with return value 0
Press any key to continue . . .
```